M. Rohaan 221-2327 Sec-E A#01

Q#01:-

Sol.

a) Categorical frequency Distribution

Trust level	Frequency	Interpretation
M (most)	21	People trust in most what they read.
H (half)	6	Few trust in one-half they read.
S (small)	2	Little one trust in small portion what they read.
A (all)	3	A very less people trust in everything what they read.

b) Relative frequency Distribution

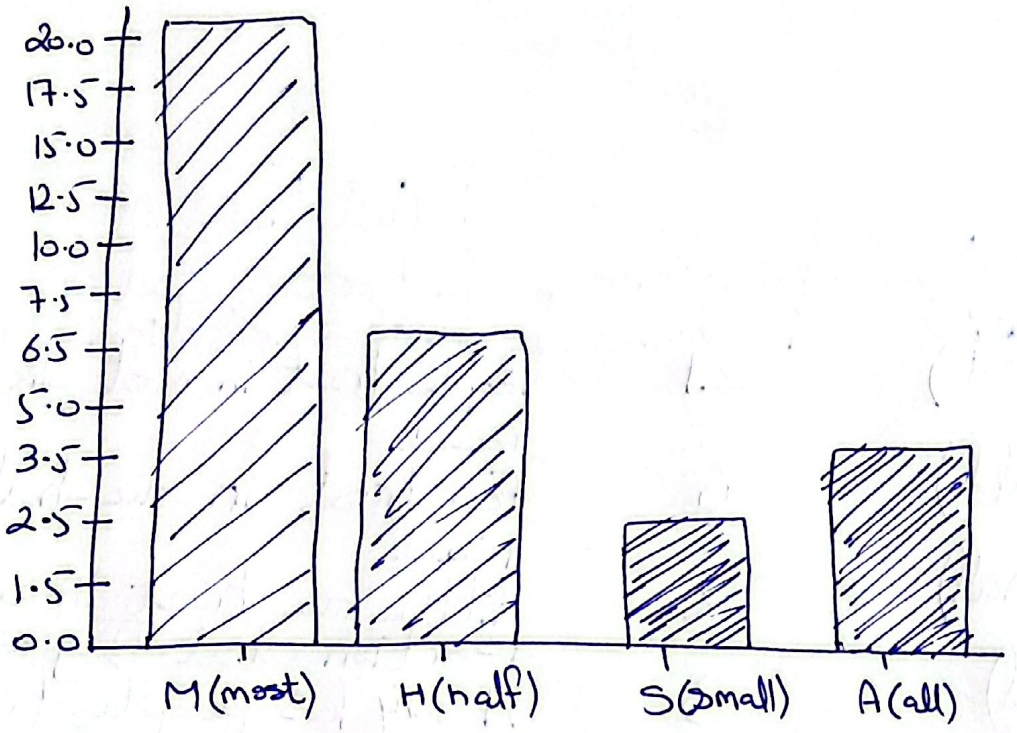
Relative frequency = frequency ÷ Total x 100

Trust Level	Frequency	Relative Frequency
M (most)	21	21/32 x 100 = 65.6%
H (half)	6	6/32 x 100 = 18.8%
S (small)	2	2/32 x 100 = 6.3%
A (all)	3	3/32 x 100 = 9.4%

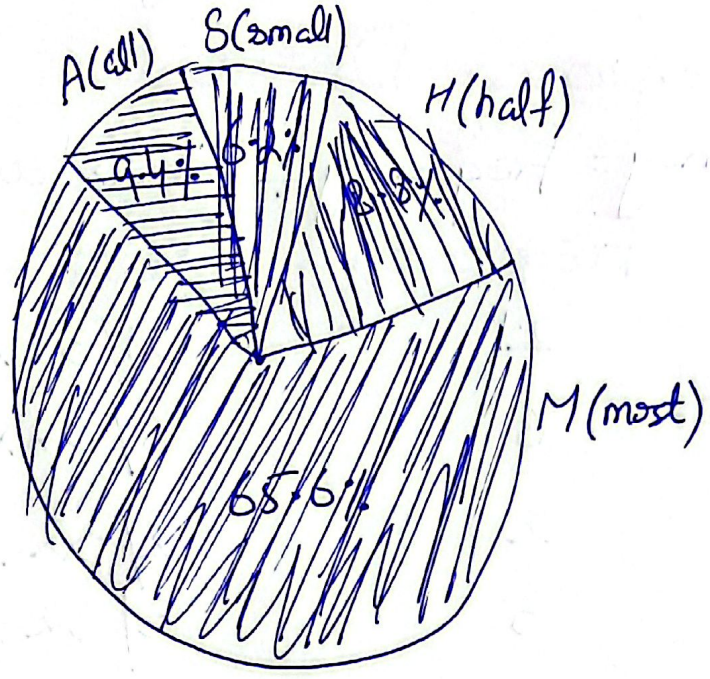
Interpretation

Maximum people exist in 'M' category means people beleive in internet information mostly. Few trust in small portion.

c) Bar Chart & Pie Chart



Bar Chart of Trust in Internet Information



Pie Chart of Trust in Internet Information

#02:-

Sol.

Min. Salary = 140

Max. Salary = 857

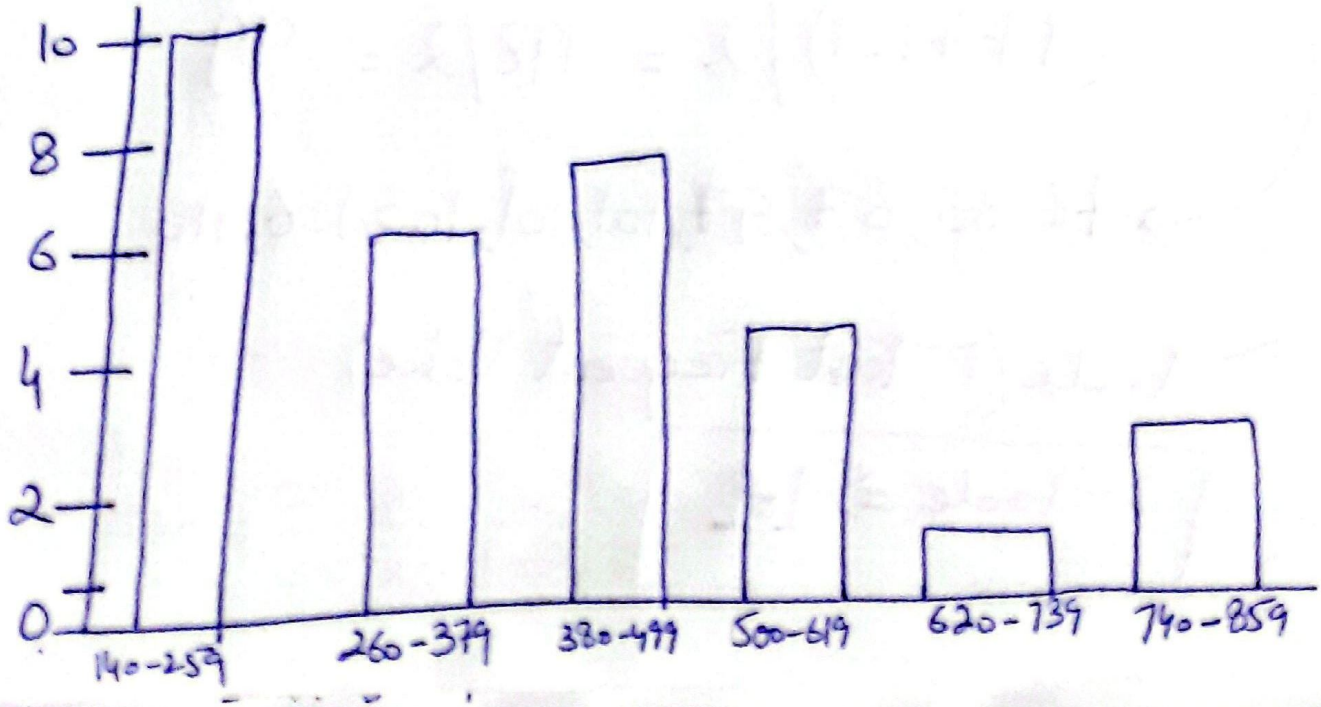
Range = 857 - 140 = 717

Class width = Range ÷ Number of classes

No. of classes = 6 (approximate, simple)

717 ÷ 6 ≈ 120

Class Interval	Frequency
140 - 259	10
260 - 379	6
380 - 499	7
500 - 619	4
620 - 739	1
740 - 859	2



Nature of Data

* Majority of Salaries 140-259k range
* Some Salaries 62ok-859k
* Histogram shape is about right-skewed.
* Mostly data is concentrated in lower salary range.

Q#03:-

Sol.

a) $\text{Mean} = \frac{88+101+97+103+110+78+87+101+80}{10}$

$= 951/10 = 95.1$

Arrange in Ascending order

Median ⇒ No. of values = 10 (even)

average of 5th and 6th values

$5^{th} = 97, \; 6^{th} = 101$

$(97+101)/2 = 198/2 = 99$

78, 80, 87, 97, 101, 101, 103, 106, 110

Mode (Most Frequent Value)

Mode ⇒ 101

Max = 110

Min = 78

Range = Max - Min

= 110 - 78

= 32

Variance, $\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{n}$

x	x-mean	$(x\text{-mean})^2$
101	5.9	34.81
97	1.9	3.61
103	7.9	62.41
110	14.9	222.01
78	-17.1	292.41
87	-8.1	65.61
101	5.9	34.81
80	-15.1	228.01
106	10.9	118.81
88	-7.1	50.41

Sum of Squares = 34.81 + 3.61 + 62.41 + 222.01 + 292.41 + 65.61 + 34.81 + 228.01 + 118.81 + 50.41

= 1112.9

Variance $= 1112.9/10 = 111.29$

Standard Deviation $= \sqrt{111.29}$

$\approx 10.5\bar{5}$

c) Rates increased by $10

* New mean = old mean $/ 10 = 95.1 \div 10 = 105.1$
* Standard Deviation doesnot change.

d) Rates become double (x2)

New mean $= 2 \times 95.1 = 190.2$

New SD $= 2 \times 10.5\bar{5} = 21.1$

e) Transformation: $Y = 5X - 3$

$Var(aX+b) = a^2 . Var(X)$, $S.D(aX+b) = |a| . SD(X)$

here $a = 5$, $b = -3$

* $Var(Y) = 5^2 \times 111.29 = 25 \times 111.29 \approx 2782.25$
* $SD(Y) = 5 \times 10.5\bar{5} \approx 52.75$

#04:-

Sol.

a) Quartiles and IQR

* Arrange in Ascending order

$n = 12$

Q_k = value at $\frac{k(n+1)}{4}$ th position

$Q1$ = 1st quartile = 25th percentile = position

$= (1 \times 13)/4 = 13/4 = 3.25$ th value

$Q1$ = 3rd value + 0.25 * (4th - 3rd) = 58 + 0.25 * (61-58

$= 58 + 0.75 = 58.75$

$Q2$ = 6.5th value = $6^{th} + 0.5 * (7^{th} - 6^{th})$ = 67 + 0.5 * (726

$= 67 + 2.5 = 69.5$ (Median)

$Q3$ = 9.75th value = $9^{th} + 0.75 * (10^{th} - 9^{th})$ = 75 + 0.75 * (78-75

$= 75 + 2.25 = 77.25$

IQR (Interquartile Range) = $Q3 - Q1 = 77.25 - 58.75$

$= 18.5$

Interpretation

→ Middle 50% students scored between 58.75 and 77.25 marks.

b) D5 and P43

* D5 (5th decile) = same as median = 69.5
* P43 (43rd percentile) = position = 43/100 x 13
  = 5.59th value

→ P43 = 5th + 0.59 * (6th − 5th) = 65 + 0.59 * (67 − 65)
= 65 + 1.18 = 66.18

Interpretation

→ 43% students scored ≤ 66.18

50% students scored ≤ 69.5

c) Identify Outliers

Outlier formula (using 1.5 x IQR)

* Lower fence = Q1 − 1.5 x IQR = 58.75 − 1.5 (18.5)
  = 58.75 − 27.75 = 31

* Upper fence = Q3 + 1.5 x IQR = 77.25 + 1.5 (18.5)
  = 77.25 + 27.75 = 105

Outliers:-

* Below 31 : 11 → Outlier
* Above 105 : None

Box and Whisker Plot

* Min(without outliers) = 57
* Q1 = 58.75
* Median (Q2) = 69.5
* Q3 = 77.25
* Max(without outliers) = 97

Outlier: 11

e) Skewness based on Box Plot

* Median closer to Q1 than Q3 → right skew
  ↳ positive skew
* Right tail longer
  ↳ higher scores spread out more than lower scores.

Q#05:-

Sol.

a) Coefficient of Variation (CV)

$$CV = \frac{\text{Standard Deviation (SD)}}{\text{Mean}} \times 100$$

Mean = Sum/n

Sum = 52+58+59+60+60+62+65+67+68+70+71+75+78+80+

= 1001

Mean = 1001 ÷ 15 ≈ 66.73

$$SD = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}}$$

Sum ≈ 1763.42

Variance = 1763.42 / 15 ≈ 117.56

SD = $\sqrt{117.56}$ ≈ 10.84

CV = (10.84 / 66.73) × 100 ≈ 16.2%

Mean = sum / n

Sum = 51 + 63 + 65 + 70 + 71 + 72 + 75 + 76 + 78 + 85 + 87 + 88 + 90 + 91 + 97

= 1159

Mean = 1159 / 15 ≈ 77.27

$\sum (x_i - \bar{x})^2 = 2187.07$

Variance = 2187.07 / 15 ≈ 145.8

SD = $\sqrt{145.8}$ ≈ 12.08

CV = (12.08 ÷ 77.27) × 100 ≈ 15.6%

Comment → Calculus CV > Linear Algebra CV = 15.6%
Calculus marks have slightly larger relative variation.

b) Box and Whisker Plot

Calculus

* Sorted: 52, 58, 59, 60, 60, 62, 65, 67, 68, 70, 71, 75, 78, 80, 96
* Median (Q2) = 8th value = 67
* Q1 = median of first 7 = (60 + 62)/2 = 61
* Q3 = median of last 7 = (75 + 78)/2 = 76.5

Min = 52, Max = 96

Sorted: 51, 63, 65, 70, 71, 72, 75, 76, 78, 85, 87, 88, 90, 91, 97

Median (Q2) = 8th value = 76

Q1 = median of first 7 = (65 + 70)/2 = 67.5

Q3 = median of last 7 = (87 + 88)/2 = 87.5

Min = 51, Max = 97

Comment:- Linear Algebra media higher → students perform better
Calculus has slightly larger spread → perform more variable